

SAYOK BOSE

EDUCATION

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Texas A&M University

College Station, TX

Bachelor of Science in Computer Science; Minor in Mathematics; Minor in Statistics (3.96)

May 2026

RELEVANT COURSEWORK

- Linear Algebra
- Differential Equations
- Engineering Calculus III
- Principles of Data Science
- Machine Learning
- Computer Organization
- Data Structures and Algorithms
- Honors Discrete Structures
- Design and Analysis of Algorithms

TECHNICAL SKILLS

Languages: Python, Java, HTML/CSS, JavaScript, C++, PHP, SQL

Technologies/Frameworks: Git, React, React Native, Ajax, Flask, CakePHP, Firebase, AWS, Azure, PyTorch, TensorFlow, Sklearn, Keras

EXPERIENCE

Texas A&M Institute for a Disaster Resilient Texas

Aug 2024 - Current

Research Assistant, Deep Learning & Geospatial Analysis

College Station, TX

- Helped design GeoSight, a deep learning framework that refines geolocation accuracy in post-disaster imagery using image similarity models (DINO, DreamSim, CLIP), reducing spatial errors by an average of 26.12 meters.
- Integrated visual-semantic search to match community-collected disaster images with reference datasets, enhancing mapping precision for tornado damage in collaboration with NOAA.
- Achieved up to 93% accuracy in top-5 image retrievals using DreamSim, improving disaster response analytics, resource allocation, and AI-driven decision systems.
- Co-authoring a publication in the *International Journal of Disaster Risk Reduction* as well as presenting at the American Meteorological Society conference to share findings with the disaster resilience research community.

Berkeley Spring Instruments

May 2024 - Aug 2024

Software Intern

Frostburg, MD

- Developed Web UIs: Designed and implemented a web user interface using CakePHP, integrated with Ajax and MySQL for dynamic data handling and improved user experience. Includes a detailed simulation of pipe flow using Three.js.
- Trained Models: Worked on developing digital twin solutions using unsupervised and generative AI with sensor data to create predictive models and automate pipe monitoring systems. Developed neural network models to predict sludge formations, analyzing results with gradient mapping and lab testing to determine optimal data forms and model types.

Sonomatic

Jun 2023 – Aug 2023

Software Engineer Intern

Houston, TX

- Lead software developer on the Joint Large Standoff Magnetometry Research Project, developing user-friendly Python software for non-intrusive geo-magnetization flux leakage measurement to detect pipeline anomalies, including efficient data scraping from magnetometers and visualizing data through graphs and 3D meshes.

PROJECTS

HowdyFinds | React Native, Firebase

May 2024

- Developed a mobile application using React Native and Firebase, providing a platform for students to buy and sell textbooks, game tickets, and other miscellaneous items. The app features a secure login system, enabling users to create accounts and manage their listings. Users can view posts, chat, and request direct messages to negotiate and finalize transactions. The integration of Firebase ensured real-time data synchronization and secure data storage, enhancing the platform's overall user experience and reliability.
- Created an LLC to formalize the HowdyFinds platform as a legal business entity, in order to scale the project and provide a robust service to the student community.

ResumeBuilder | Python, HTML, JavaScript

March 2024

- Developed an innovative Flask-based application that leverages machine learning to analyze resumes and automate the job application process. The program employs web scraping techniques to extract key information from job postings, then utilizes Azure OpenAI Service to generate text for tailored, formatted resumes for each application.

NBA Prediction Model | Python

Feb 2024

- Utilized web scraping techniques to gather comprehensive NBA game data, extracting player and team statistics for analysis. Leveraging Python and data manipulation libraries, I engineered features such as player performance metrics and opponent statistics to develop predictive models for forecasting player performance.

AWARDS

Brown Scholar

Feb 2022-Current

Craig and Galen Brown Foundation

Texas A&M University

- Awarded a full-ride scholarship from the Craig and Galen Brown Foundation and National Merit Finalist President Endowed Scholarship for outstanding academic achievement, National Merit Finalist status, leadership, community service, and communication skills..

INVOLVEMENT

Hindu Heritage Youth Camp

Aug 2021 - Current